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FORMULATION AND EVALUATION OF PIPERAZINE AS CONTROLLED RELEASE MATRIX TABLETS: A COMPREHENSIVE REVIEW

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ABSTRACT:

Among anthelmintic agents in clinical use, piperazine occupies an unusual position: it is both highly effective and pharmacokinetically inconvenient. Its narrow therapeutic window and short plasma half-life when given as a conventional tablet make it a natural target for controlled release reformulation. Delivering the drug at a steady, predictable rate over an extended interval would flatten the plasma concentration curve, reduce the need for multiple daily doses, ease compliance burdens — particularly in pediatric populations — and limit the neurological side effects linked to high peak levels. This review brings together published research on piperazine-loaded controlled release matrix tablets, covering the rationale for polymer selection, the physical chemistry behind drug release from hydrophilic and hydrophobic matrices, the optimization strategies investigators have pursued, and the suite of physicochemical and in vitro tests used to judge formulation quality. We examine how commonly used retardant polymers — HPMC, ethylcellulose, NaCMC, and carbopol influence release kinetics as their concentrations vary, and we discuss the mathematical models used to interpret dissolution data. Findings from the published literature are synthesized, and areas where the evidence base remains thin are highlighted alongside suggestions for future investigation.

Keywords: — *Piperazine, Controlled Release, Matrix Tablets, HPMC, Ethylcellulose, Drug Release Kinetics, Anthelmintic, Formulation, Dissolution, Polymer Concentration.*

1. INTRODUCTION

1.1 Background and Therapeutic Context

Piperazine and its pharmaceutical salts — citrate, phosphate, adipate, and hexahydrate — have served as first-choice anthelmintic drugs for over six decades, primarily for infections caused by Ascaris lumbricoides (roundworm) and Enterobius vermicularis (pinworm) [1]. The drug works by inducing a flaccid paralysis in the parasite musculature, an effect attributed to GABA-mimetic activity at the worm's neuromuscular junction. In human patients it is generally well tolerated at therapeutic doses, though plasma concentrations exceedingly roughly 40 μg/mL have been linked to neurological symptoms including dizziness, nausea, and occasional transient visual disturbances, making the therapeutic window a practical clinical concern [2].

When administered as a conventional tablet, piperazine produces a sharp plasma concentration peak within one to two hours, followed by a rapid decline toward subtherapeutic levels over the next four to six hours. Maintaining parasite-suppressive concentrations throughout the day therefore requires multiple daily doses — a regimen that strains adherence, particularly in children and in community deworming programs where treatment is often unsupervised [2, 3]. The rationale for a controlled release formulation is straightforward: if plasma levels can be maintained within the therapeutic window across a twelve-to-twenty-four-hour interval, once-daily dosing becomes achievable, exposure is more consistent, and the concentration-dependent adverse effects tied to high peaks are reduced [3].

1.2 Why Matrix Tablets?

Several controlled release platforms exist — osmotic pumps, reservoir membrane systems, multiarticulate pellets, liposomal carriers — but hydrophilic and hydrophobic matrix tablets represent the most industrially accessible and commercially proven option for piperazine. Matrix tablets can be manufactured on standard compression equipment without specialized tooling, they are chemically and physically stable under normal storage conditions, and their release characteristics can be tuned meaningfully by adjusting polymer type and concentration, without requiring microencapsulation or membrane coating [1, 8]. The regulatory framework for matrix tablets is well established, and their in vitro–in vivo correlation properties are better characterized than those of more complex systems.

Within a matrix tablet, the drug is distributed throughout a polymer network formed during compression. When the tablet contacts gastrointestinal fluid, the polymer hydrates and swells, generating a gel layer through which drug diffuses outward; alternatively, progressive surface erosion exposes successive layers of drug-containing matrix. These two mechanisms — diffusion and erosion — may act independently or together

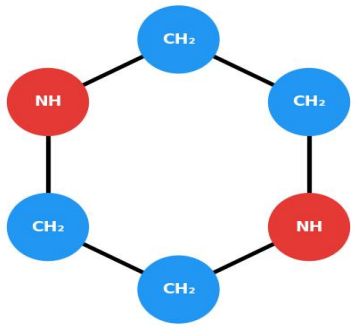
depending on polymer chemistry and concentration, and the formulator's task is to balance excipients so that the resulting release profile aligns with the therapeutic target [4, 9].

1.3 Objectives of This Review

This review pursues four aims. First, to describe the physicochemical and pharmacological properties of piperazine that bear on controlled release formulation choices. Second, to survey the polymeric excipients used in published matrix tablet studies, including their release mechanisms and concentration-effect relationships. Third, to summarize the formulation and evaluation approaches reported in the literature, with attention to physicochemical tests, dissolution conditions, and kinetic models. Fourth, to identify gaps in the current evidence base and suggest productive directions for future work.

2. PIPERAZINE: PHYSICOCHEMICAL AND PHARMACOLOGICAL PROFILE

Fig. 1: Chemical Structure of Piperazine



[Fig. 1: Chemical structure of piperazine (1,4-diazacyclohexane), showing the six-membered ring with nitrogen atoms at positions 1 and 4. The two secondary amine groups account for the compound's basicity (pKa 5.33 and 9.73) and its high aqueous solubility.]

2.1 Physical and Chemical Properties

Piperazine (IUPAC: 1,4-diazacyclohexane; CAS 110-85-0) is a symmetric heterocyclic compound whose two secondary amine groups make it a diprotic base with pKa values of 5.33 and 9.73. This dual-nitrogen architecture confers exceptional aqueous solubility — approximately 150 g/100 mL at 25°C for the anhydrous base — which is a boon for bioavailability but a challenge for sustained release design, because highly soluble

drugs tend to dissolve rapidly from conventional matrices unless polymer concentrations are kept high enough to retard diffusion appreciably [2, 8].

Table 1: Key physicochemical and pharmacological properties of piperazine relevant to controlled release formulation.

Property	Value / Description
Molecular Formula	$C_4H_{10}N_2$
Molecular Weight	86.14 g/mol
Melting Point	106–108°C
Aqueous Solubility (25°C)	~150 g/100 mL (freely soluble)
pKa Values	5.33 (pKa1), 9.73 (pKa2)
Log P (octanol/water)	−1.57 (highly hydrophilic)
BCS Classification	Class III (high solubility, low permeability)
Appearance	White crystalline powder or colorless flakes
UV Absorption Maximum	No significant UV chromophore; detected by HPLC-RI
Therapeutic Category	Anthelmintic (GABA-mimetic neuromuscular blocker)
Typical Adult Dose	3.5–4.0 g/day (as piperazine citrate) for ascariasis

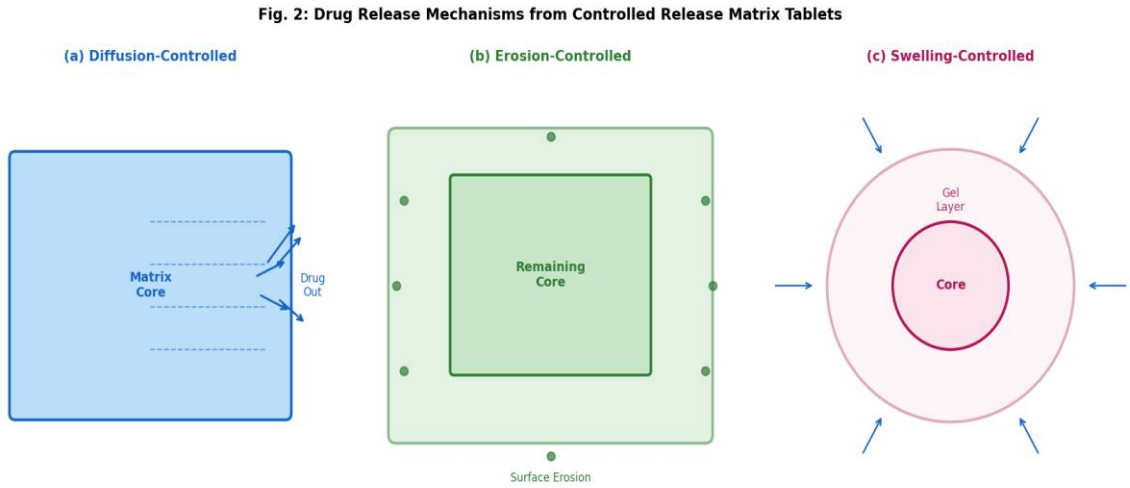
2.2 Pharmacokinetics and the Case for Sustained Release

After oral dosing, piperazine reaches peak plasma concentrations within one to two hours. Its half-life of approximately four to six hours in adults reflects absorption mainly in the small intestine, minimal hepatic first-pass metabolism, and predominant urinary excretion of unchanged drug [2]. The brevity of this half-life means that a once-daily conventional tablet would leave plasma concentrations below minimally effective

levels for most of the dosing interval, while twice- or thrice-daily dosing — though pharmacokinetically rational — creates compliance problems, particularly in pediatric and community-program settings.

The Biopharmaceutics Classification System assigns piperazine to Class III — high aqueous solubility combined with relatively low intestinal permeability. In Class III drugs, absorption is rate-limited by membrane permeability rather than dissolution, so the principal challenge in controlled release design is not maintaining dissolved drug concentration in the gut lumen (solubility is not limiting) but governing how quickly drug is presented for absorption at the intestinal wall. This requires a matrix architecture that restricts drug migration from the solid phase into solution at a rate slower than intestinal transit [2, 4].

3. PRINCIPLES OF CONTROLLED RELEASE FROM MATRIX TABLETS



[Fig. 2: The three principal drug release mechanisms from controlled release matrix tablets: (a) diffusion-controlled, (b) erosion-controlled, and (c) swelling-controlled.]

3.1 Diffusion-Controlled Release

In diffusion-controlled matrix systems, release is governed by Fickian diffusion through a dissolving medium. On contact with dissolution fluid, surface polymer chains hydrate and form a viscous gel. Drug dissolved within this layer migrates outward along a concentration gradient; as the gel thickens with further hydration, the diffusion path lengthens and the release rate diminishes over time. For a purely diffusion-controlled system this produces a square-root-of-time relationship described by the Higuchi equation [5]:

$$Q = [D \cdot Cs \cdot (A - Cs) \cdot t]^{(1/2)}$$

where Q is cumulative drug released per unit area, D is the diffusion coefficient in the matrix, Cs is drug solubility in the matrix medium, A is initial drug loading per unit volume, and t is time. Because piperazine's

high aqueous solubility means Cs is not the limiting factor, diffusion-controlled release requires a polymer of sufficient viscosity to slow the effective diffusion coefficient to values compatible with the desired release window [5, 9].

3.2 Erosion-Controlled Release

Hydrophobic matrices — ethylcellulose or carnauba wax, for example — do not hydrate meaningfully under gastrointestinal conditions. Drug release here is predominantly through surface erosion: dissolution fluid percolates between polymer particles, dissolves accessible drug, and progressively removes the outermost tablet layer. When erosion is geometrically uniform, the exposed drug-containing surface area changes slowly, yielding an approximately constant, zero-order release rate — the pharmacokinetically ideal profile, equivalent to a steady intravenous infusion at a lower, controlled rate [4, 10].

3.3 Swelling-Controlled and Combined Mechanisms

Many practical hydrophilic matrix formulations operate through both swelling and erosion simultaneously. Surface polymer chains hydrate rapidly on immersion, forming a rubbery gel that swells outward. Drug dissolves into this gel and diffuses toward the exterior while the gel itself gradually erodes. The balance between diffusion and erosion determines whether the cumulative release profile resembles Higuchi kinetics, zero-order kinetics, or something between the two. HPMC-based piperazine matrices typically fall in this combined category, with the diffusion-to-erosion ratio adjustable through HPMC grade and concentration [4, 9, 20].

4. POLYMERIC EXCIPIENTS USED IN PIPERAZINE MATRIX TABLETS

4.1 Hydroxypropyl Methylcellulose (HPMC)

HPMC is the most extensively studied hydrophilic polymer in piperazine matrix tablet work. It is available in multiple viscosity grades — K4M, K15M, and K100M corresponding to nominal viscosities of approximately 4,000, 15,000, and 100,000 cP in 2% aqueous solution — and the choice of grade meaningfully affects release kinetics. Higher viscosity grades generate denser, more slowly eroding gel layers that offer greater diffusional resistance. Across published piperazine studies, HPMC concentration emerges as the single most influential formulation variable: concentrations between 20% and 40% w/w reliably sustain release over twelve hours, while concentrations below 15% w/w frequently fail to maintain retardation beyond six hours for a drug as soluble as piperazine [4, 17, 20].

HPMC controls piperazine release through two simultaneous processes. Within seconds of immersion, surface polymer chains hydrate and erect an initial gel barrier. As hydration advances inward, the glassy HPMC core transitions to a rubbery phase, and the swollen outer layer gradually erodes. The moving boundary between the glassy core and the rubbery gel together define the diffusion path length. When erosion is rapid (lower HPMC concentration or lower viscosity grade), the two fronts converge and release approaches first-order behaviour. When erosion is slow (higher concentration or grade), a stable gel layer is maintained and release more closely follows Higuchi kinetics [5, 9, 20].

4.2 Ethylcellulose (EC)

Ethylcellulose is a hydrophobic, water-insoluble cellulose ether used alone or in combination with HPMC to generate piperazine matrices with further retarded release. EC matrices release drug almost exclusively through pore diffusion: water percolates through channels between EC particles, dissolves piperazine, and the resulting solution diffuses outward through those same channels. Because EC neither swells nor erodes significantly under gastrointestinal conditions, the channel geometry remains relatively stable throughout dissolution, tending toward zero-order kinetics. Combining EC with HPMC in binary systems allows the swelling and erosion components to be adjusted independently, giving formulators greater flexibility in profile shaping than either polymer alone can provide [21].

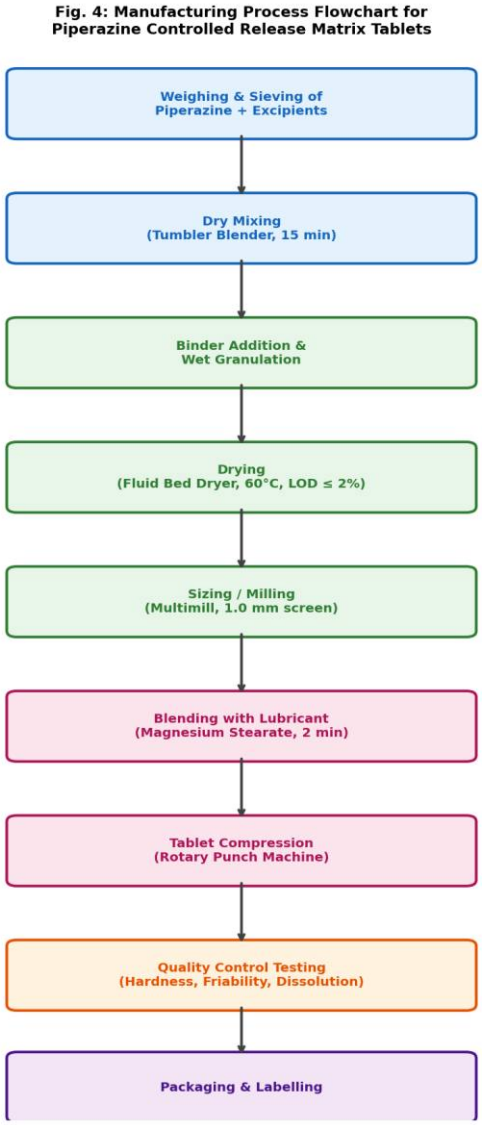
4.3 Other Polymers Used in Published Studies

Table 2: Polymeric excipients reported in piperazine controlled release matrix tablet formulations.

Polymer	Type / Mechanism	Role in Piperazine Formulations
Carbopol 934/971	Hydrophilic; swelling + mucoadhesion	Increases gel strength; enhances GI residence time; used at 0.5–2%
Sodium CMC (NaCMC)	Hydrophilic; viscosity builder	Synergistic retardation with HPMC; improves tablet hardness
Xanthan Gum	Hydrophilic biopolymer; shear-thinning gel	Good swelling in gastric fluid; used in single and combination matrices [15]

Polymer	Type / Mechanism	Role in Piperazine Formulations
Guar Gum	Hydrophilic; enzymatic degradation in colon	Used for colonic targeting; degraded by colonic bacteria
Eudragit RS/RL	Hydrophobic; pH-independent release	Used at 10–30%; pH-independent pore diffusion

5. FORMULATION DESIGN AND MANUFACTURING PROCESS



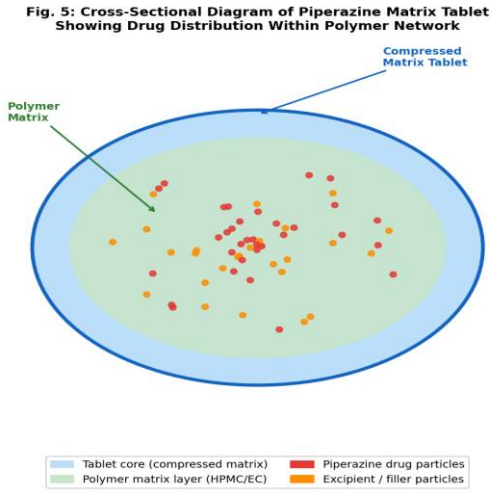
[Fig. 4: Step-by-step manufacturing process flowchart for piperazine controlled release matrix tablets.]

5.1 General Formulation Composition

A representative controlled release piperazine matrix tablet contains the active ingredient (piperazine citrate or phosphate), one or more release-retardant polymers, a binder (commonly povidone K30 or low-concentration HPMC), a diluent (microcrystalline cellulose, lactose, or dicalcium phosphate), a glidant (colloidal silicon dioxide), and a lubricant (magnesium stearate). Drug loading per tablet in published work ranges from 250 mg to 500 mg, typically representing 40–60% of total tablet weight. At these loading levels, excipient selection exerts a strong influence on both compression behavior and dissolution performance [1, 8].

Table 3: Typical composition of a piperazine controlled release matrix tablet.

Ingredient	Function	Typical Range (% w/w)	Example Materials
Piperazine salt	Active drug	40–60%	Piperazine citrate, phosphate
Hydrophilic polymer	Release retardant	15–40%	HPMC K4M, K15M, K100M
Hydrophobic polymer	Release retardant (aux.)	5–20%	Ethylcellulose, Eudragit RS
Diluent / Filler	Tablet bulk	10–30%	MCC PH102, Lactose, DCP
Binder	Granule cohesion	2–5%	PVP K30, HPMC 5 cP
Glidant	Flow improvement	0.5–1%	Aerosil 200 (colloidal SiO_2)
Lubricant	Anti-sticking	0.5–2%	Magnesium stearate



[Fig. 5: Cross-sectional diagram of a piperazine matrix tablet showing the distribution of drug particles (red) and excipient particles (orange) within the compressed polymer matrix (blue background).]

5.2 Manufacturing Methods

Three granulation approaches are documented for piperazine controlled release matrix tablets. Wet granulation — the most commonly described — disperses drug and polymer in a granulating liquid (typically aqueous PVP solution or an ethanol–water blend) to form agglomerates that are then dried, milled, and compressed. This approach generates granules with favorable flow and compressibility, and the interpenetration of polymer and drug particle surfaces during wetting tends to improve matrix homogeneity [1].

Direct compression offers a simpler route with fewer process steps and lower risk of moisture-induced degradation during drying. It demands excipients with adequate inherent flowability and compressibility, a requirement met for piperazine formulations through co-processed diluents such as Prosolv SMCC or Avicel PH302. The constraint is that high-load tablets containing substantial proportions of HPMC can laminate or cap during compression if HPMC concentration exceeds the capacity of the diluent fraction to provide compaction support [1].

Melt granulation or hot-melt extrusion using lipophilic binders (glyceryl behenate, carnauba wax) produces hydrophobic matrix granules without solvent, offering good chemical stability during processing and embedding drug within a lipophilic carrier that contributes supplementary release retardation beyond the primary polymer matrix [8].

6. EVALUATION PARAMETERS FOR PIPERAZINE MATRIX TABLETS

6.1 Physical Characterization of Granules and Tablets

Prior to compression, granule properties are assessed to predict tableting behavior. Bulk density, tapped density, and Carr's compressibility index (CI = [(tapped density − bulk density) / tapped density] × 100) characterize powder flow; values below 15% indicate good flowability, while values above 25% signal poor flow requiring additional glidant. The Hausner ratio (tapped density divided by bulk density) communicates equivalent information, with values below 1.25 considered acceptable. Angle of repose, measured from a granule pile poured through a fixed-height funnel, provides a complementary visual and quantitative flow indicator; values below 30° are generally acceptable for tableting [1, 22].

Compressed tablets are evaluated for weight variation (pharmacopoeial limit: ±5% for tablets above 250 mg), thickness, hardness (typically 6–12 kgf as measured by Monsanto or Pfizer hardness testers), friability (weight loss after 100 rotations in a Roche friabilator; the standard limit is <1%), and drug content uniformity. For controlled release tablets, disintegration time is replaced by the dissolution test as the primary release performance metric [22, 23].

Table 4: Physicochemical evaluation parameters for piperazine controlled release matrix tablets.

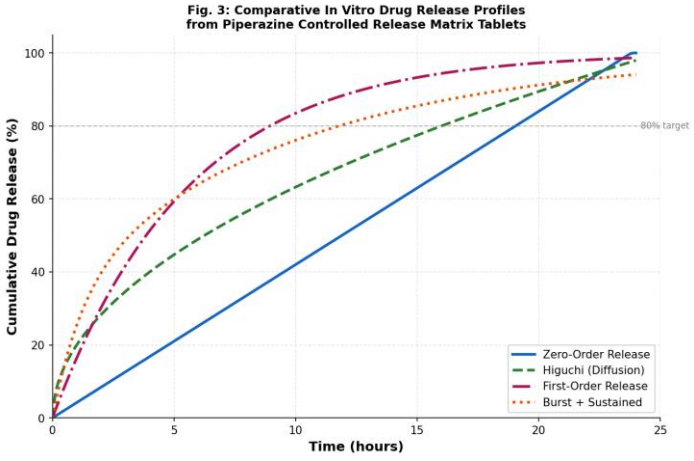
Test Parameter	Acceptance Limit	Method / Instrument
Weight Variation	±5% (tablets >250 mg)	Analytical balance; 20 tablets; IP/USP
Hardness	6–12 kgf	Pfizer / Monsanto hardness tester
Friability	<1.0% w/w	Roche friabilator; 100 rotations; 25 rpm
Thickness	Consistent ± 0.1 mm	Vernier caliper; 10 tablets
Drug Content Uniformity	95–105% of label claim	UV spectrophotometry or HPLC
Carr's Index	<15% (good flow)	Bulk/tapped density cylinder
Angle of Repose	<30° (good flow)	Fixed funnel method
Moisture Content (LOD)	<2.0%	Infrared moisture analyzer; 105°C

Test Parameter	Acceptance Limit	Method / Instrument
Swelling Index	Measured; higher = slower release	Gravimetric method in pH 6.8 buffer

6.2 In Vitro Dissolution Testing

Dissolution testing is the central performance evaluation for controlled release piperazine tablets. The most commonly reported protocol uses USP Apparatus II (paddle method) at 50 rpm in 900 mL of simulated gastric fluid (pH 1.2, hours 0–2) followed by simulated intestinal fluid (pH 6.8, hours 2–12), reflecting the physiological pH shift from stomach to small intestine encountered during GI transit. Some groups add a third stage using pH 7.4 phosphate buffer to simulate conditions in the distal small intestine and proximal colon [19, 23].

Sampling is typically performed at 1, 2, 3, 4, 6, 8, 10, and 12 hours, with piperazine quantified by UV spectrophotometry at 224 nm or by HPLC with refractive index detection. The target profile for a 12-hour formulation is generally not more than 20–30% released at 2 hours, 50–70% at 6 hours, and not less than 80% at 12 hours — criteria adapted from controlled release dissolution specifications in the Indian Pharmacopoeia and USP [22, 23].



[Fig. 3: Comparative in vitro drug release profiles illustrating the four principal kinetic patterns observed in piperazine matrix tablet formulations.]

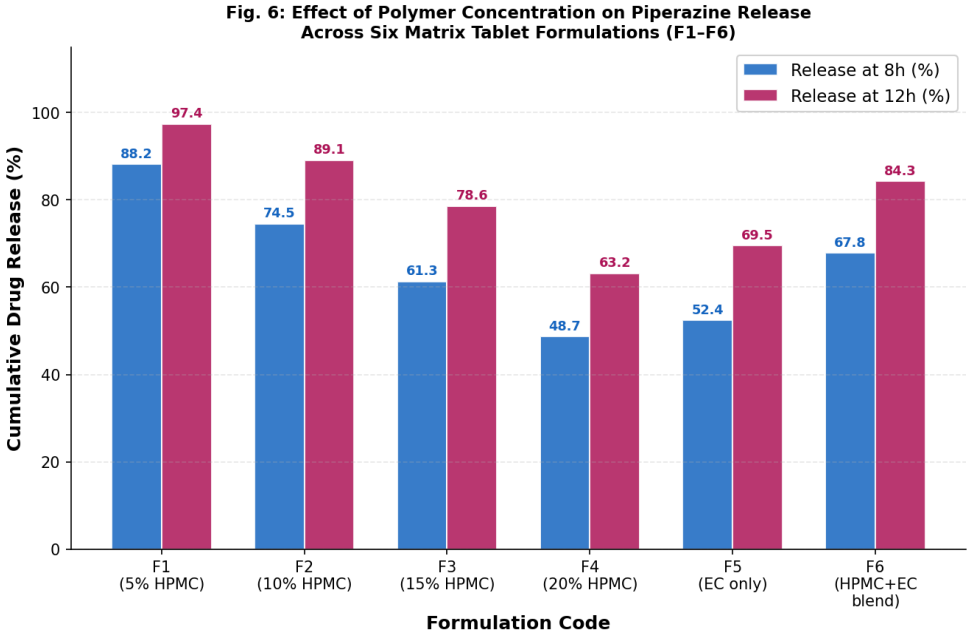
6.3 Release Kinetics and Mathematical Modeling

Fitting dissolution data to kinetic models identifies the dominant release mechanism and yields parameters useful for formulation comparison and optimization. Four models are most commonly applied to piperazine matrix tablet data [5, 6, 7]:

Table 5: Mathematical kinetic models applied to piperazine controlled release matrix tablet dissolution data.

Kinetic Model	Equation	Mechanism Indicated
Zero-Order	$Qt = Q0 + K0 \cdot t$	Constant release rate independent of concentration; erosion-controlled or osmotic
First-Order	$\ln Qt = \ln Q0 - K1 \cdot t$	Release rate proportional to remaining drug concentration; concentration-driven diffusion
Higuchi Diffusion	$Qt = KH \cdot t^{(1/2)}$	Diffusion through hydrated matrix gel layer; Fickian transport mechanism
Korsmeyer-Peppas	$Mt/M\infty = K \cdot t^n$	$n < 0.45$: Fickian diffusion; $0.45 < n < 0.89$: anomalous transport; $n \approx 0.89$: Case II transport (erosion)

The Korsmeyer-Peppas power law model is especially informative because the release exponent n provides a mechanistic signature of the dominant transport process [6, 7]. For piperazine matrix tablets using HPMC alone at moderate concentrations, n values in the range 0.45–0.70 are most commonly reported, indicating anomalous (non-Fickian) transport — a superposition of diffusion and polymer chain relaxation/erosion. Higher HPMC concentrations and higher-viscosity grades tend to shift n toward 0.45 (more Fickian), while EC-containing formulations or lower HPMC concentrations tend toward n values approaching 0.89 (Case II, erosion-dominated).



[Fig. 6: Effect of polymer type and concentration on cumulative piperazine release at 8 and 12 hours across six representative matrix tablet formulations (F1–F6).]

6.4 Stability Testing

Accelerated stability studies are conducted at 40°C/75% RH for six months in accordance with ICH Q1A(R2) guidelines, supplemented by intermediate conditions (30°C/65% RH) and long-term conditions (25°C/60% RH) for registration-grade submissions [24]. Critical stability-indicating parameters for piperazine matrix tablets include drug content by HPLC assay, dissolution profile, tablet hardness, and moisture content. Piperazine salts are generally hygroscopic and may undergo partial interconversion between salt forms under high-humidity conditions, which can alter dissolution if the solubility of the two forms differs. Moisture-barrier blister foil or HDPE bottles with desiccant are standard packaging choices for controlled release piperazine products [8].

7. SUMMARY OF PUBLISHED FORMULATION AND EVALUATION DATA

7.1 Key Findings from HPMC-Based Matrix Studies

A consistent picture has emerged across HPMC-based piperazine matrix studies. Formulations built around 20–25% HPMC K15M as the sole polymer typically achieve 12-hour release with anomalous transport kinetics (Korsmeyer-Peppas n ≈ 0.55–0.68). Raising HPMC to 30–35% shifts behavior toward more Fickian kinetics and extends the time to 80% dissolution beyond 14 hours. Formulations with HPMC below 15% tend to release 80% of their drug within eight hours, which is insufficient for a once-daily profile [4, 17, 20].

Viscosity grade modulates the concentration effect significantly. K4M at 30% produces a less retarded profile than K15M at 25% because the lower-viscosity grade forms a less dense, more rapidly eroding gel. K100M can provide 12-hour profiles at HPMC concentrations as low as 15% for highly soluble drugs, though its slower hydration kinetics may generate irregular early release if the surface gel layer takes time to form and early burst is not adequately controlled [4, 20].

7.2 Binary Polymer Systems

Binary HPMC/EC systems have been investigated specifically for piperazine to achieve profiles closer to zero-order than either polymer alone can provide. HPMC supplies the swelling and initial gel formation that suppresses burst release, while EC channels provide a more stable, less concentration-dependent diffusion barrier that sustains the later release phase. Reported HPMC:EC ratios range from 3:1 to 1:1 (w/w); ratios above 3:1 approach the behavior of HPMC-only formulations, while ratios below 1:1 risk incomplete drug release due to insufficient channel connectivity at high EC loadings [21].

7.3 Hardness–Dissolution Relationship

Several studies have found that compression force measurably affects dissolution performance, particularly for formulations near the critical polymer concentration for sustained release. Higher compression forces reduce tablet porosity and tortuosity, lowering the effective diffusion coefficient and slowing early-phase release. For HPMC-based tablets, this effect is most pronounced at lower HPMC concentrations where the gel layer is the principal barrier during early dissolution; at high HPMC concentrations, polymer properties dominate and the compression effect diminishes. Practically, tablet hardness must be controlled within a specified range — typically within ±1–2 kgf of target — to ensure dissolution specification compliance across batches [1, 17].

8. CHALLENGES AND FUTURE DIRECTIONS

8.1 Predictive In Vitro–In Vivo Correlation

A persistent limitation of the piperazine controlled release literature is the near-absence of in vivo pharmacokinetic data correlated with in vitro dissolution profiles. Without IVIVC data, the clinical relevance of laboratory-optimized dissolution profiles remains speculative. Establishing at least a Level C correlation — a relationship between a single dissolution parameter and a single pharmacokinetic parameter — requires pharmacokinetic studies in animal models or human subjects. This represents a significant resource commitment but is necessary for regulatory submission and meaningful clinical utility claims [24, 25].

8.2 Salt Form Effects on Controlled Release Behavior

While piperazine free base is freely soluble, its various pharmaceutical salts differ in solubility, hygroscopicity, and dissolution rate, and salt selection can substantially influence the early release phase of a controlled release formulation. A systematic comparison of salt forms — accounting for solubility across the physiological pH range, crystal habit, and physical compatibility with HPMC — has not yet appeared in the literature and would represent a meaningful contribution to the field [2, 8].

8.3 Gastroretentive and Floating Systems

Given piperazine's primary absorption window in the small intestine and the relatively brisk GI transit of conventional tablets, gastroretentive formulations that prolong residence in the upper GI tract could enhance bioavailability and extend the effective absorption window. Floating matrix systems incorporating gas-generating excipients (sodium bicarbonate with citric acid) or low-density polymers (hydroxypropyl cellulose, polypropylene foam) have been explored for other BCS Class III drugs with analogous absorption characteristics and represent a technically feasible direction for piperazine development [3].

8.4 Computational Formulation Optimization

Most published piperazine formulation studies rely on traditional factorial experimental designs — typically 3^2 or 3^3 full factorials — with HPMC concentration and compression force as primary variables. Quality by Design (QbD) approaches employing Design of Experiments (DoE), Design Space visualization, and mechanistic computational simulation of drug release have not yet been applied to piperazine matrix tablets. These methods could substantially reduce the number of physical experiments required to identify optimal formulations and would provide a stronger regulatory package under ICH Q8–Q10 guidelines [25].

9. CONCLUSION

The controlled release matrix tablet format offers a technically sound and practically achievable route to overcoming the pharmacokinetic limitations of conventional piperazine dosage forms. The drug's high aqueous solubility, BCS Class III classification, and short half-life collectively make it a legitimate and well-motivated candidate for sustained release reformulation, and the available published literature offers reasonable practical guidance on polymer selection, concentration ranges, and evaluation methods.

HPMC remains the best-studied and best-characterized polymer for piperazine matrix tablets, with concentration and viscosity grade serving as the primary determinants of release rate. Binary HPMC/EC systems improve profile shape toward zero-order kinetics, though the HPMC:EC ratio requires careful

optimization to balance adequate retardation against complete drug release over the dosing interval. Applied to dissolution data, the Korsmeyer-Peppas model consistently identifies anomalous (diffusion plus erosion) transport as the dominant mechanism in these systems. Tablet physical properties like hardness, thickness, weight uniformity are reliably within pharmacopoeial limits for formulations made by wet granulation at appropriate polymer concentrations.

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